RF Energy Harvesting via Monopole Antenna-Coupled Rectifier

hardware implementation, circuit explanation and efficiency

1. Hardware implementation

The prototype is built on a copper-clad board that acts as the ground plane. Components are soldered directly to the circuit with short leads, which keeps stray inductance and capacitance low at 2.4 GHz. A wire monopole of about 3 cm (a quarter wavelength at 2.44 GHz is 3.07 cm) feeds the circuit at the top of the board.

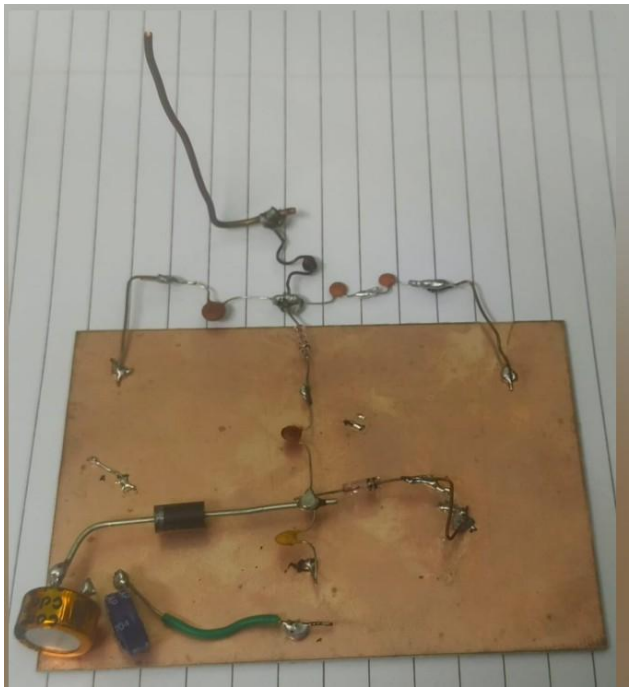


Figure 1. Prototype on a copper-clad ground plane (wire monopole at the top, storage supercapacitor bottom left).

What the photograph shows (identified by appearance):

- Wire monopole antenna at the top, soldered to the matching network.
- Small orange ceramic disc capacitors for the picofarad matching and coupling parts.
- Small glass-body diodes and one larger axial diode with a cathode band.
- A yellow supercapacitor (bottom left) used as the energy store, and a blue capacitor marked 104 (100 nF).

Note on the diodes. The leaded diodes visible in this build are not the SMS7630 devices used in the final simulated design (Part 1), which is a surface-mount part. The photograph therefore shows the first build of the circuit family, and the SMS7630 version needs to be assembled and measured before its results can be claimed for the hardware.

2. How the circuit works

The harvester is a chain of four stages. Each stage and its parts are described below, using the values from the final design.

Stage	Parts	What it does
Capture	3 cm wire monopole	Turns the incident 2.44 GHz field into an RF voltage and current. Over a ground plane a quarter-wave monopole has an impedance of about 36 Ω, close to the 40 + j10 Ω used in the project calculations.
Match	C1 0.7 pF, L1 18.5 nH	An L-section that transforms the high impedance of the rectifier input down to the antenna impedance, so that power is absorbed instead of reflected. It also raises the voltage across the diodes, which is what lets them conduct at microwatt power.
Rectify	C2 10 pF, D1, D2 (SMS7630), C4 100 pF	A Villard voltage doubler. C2 couples the RF and blocks DC; D1 clamps the waveform so its negative peaks sit near 0 V; D2 charges C4 to about twice the RF peak. SMS7630 diodes have 0.14 pF junction capacitance and a low turn-on voltage, which keeps losses small at tiny signal levels.
Store and use	C4, supercapacitor, load	C4 smooths the rectified voltage. A supercapacitor then accumulates the energy ($E = \frac{1}{2}CV^2$) and supplies a low-power load in short bursts. As it charges, its voltage changes the load the rectifier sees.

Reactances of the key parts at 2.44 GHz:

Part	Value	Reactance
L1	18.5 nH	+j283.6 Ω
C1	0.7 pF	−j93.2 Ω
C2	10 pF	−j6.5 Ω
C4	100 pF	−j0.65 Ω (a short at RF)

3. Circuit efficiency

The RF-to-DC efficiency is **PCE = P_dc / P_av**, where P_av = 10^(Pdbm/10) mW is the power available from the antenna and $P_{dc} = V_{out}^2 / R_L$ for a resistive load. The overall efficiency is the product of the antenna efficiency, the matching efficiency $(1 - |\Gamma|^2)$ and the rectifier efficiency. The simulated PCE below includes the matching loss because it is measured against the available source power.

P_av (dBm)	P_av (µW)	V_out (mV), LTspice	P_dc (µW)	PCE (%), LTspice
−30	1.0	≈ 18	≈ 0.03	≈ 3.2
−25	3.2	≈ 55	≈ 0.30	≈ 9.6
−20	10	≈ 140	≈ 1.96	≈ 19.6
−15	31.6	≈ 330	≈ 10.9	≈ 34.4
−10	100	≈ 660	≈ 43.6	≈ 43.6

Values are read from the LTspice plots (Part 1, Figure 2) with R_L = 10 kΩ. With a realistic inductor (Q = 50) the ngspice re-simulation gives 4.5, 11.5, 23.3, 37.0 and 45.9 % at the same five levels.

4. What input power to expect

For a Wi-Fi router at 100 mW EIRP, a 2.15 dBi receiving antenna and free-space propagation, the Friis equation gives the received power below. Together with the table above this shows the efficiency to expect at each distance.

Distance	Received power	Level	Expected PCE (simulation)
1 m	≈ 16 µW	≈ −18 dBm	≈ 25–35 %
2 m	≈ 3.9 µW	≈ −24 dBm	≈ 11–14 %

Distance	Received power	Level	Expected PCE (simulation)
5 m	≈ 0.63 µW	≈ −32 dBm	≈ 3 % or less

Expected values are interpolated between the simulated points (including the Q = 50 case) and assume free-space propagation; real rooms and a real antenna will differ. 30 % efficiency therefore needs about −17 dBm, roughly 1 m from a router or a higher-gain antenna.